Report

Assignment 5

Data Analytics E0 – 259

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The objective of the assignment is to upload data files, applying a search algorithm to match reads to the reference sequence, getting red exon and green exon counts and likelihood of the occurrence of a configuration

Implementation Summary

To find the matches to the reference sequence exhaustive search could not be done because of the time and space complexity (O(nm)). In this case the time complexity reaches 150 trillion which is too inefficient to calculate.
To address this, **the Burrows-Wheeler Transform (BWT)** is used to enable backward search, reducing the time complexity to $O(\delta)$ by leveraging rank and select operations, where delta represents the interval at which data is sampled. BWT is applied to the reference sequence, enabling backward searches with rank and select operations. These operations utilize milestone markers (delta intervals) that store cumulative character counts, allowing targeted searching within the compressed BWT data. **Rank** counts occurrences of each character up to a specific position. **Select** locates the position of a character at a given rank. Together, these operations allow a rapid mapping of reads to their potential positions within the genome. Reads of gene 1 and 6 cannot be assigned ambiguously as these two exons are identical and there is always sufficient number of transcripts present, hence are not necessary for colour blindness detection. Rest of the 2, 3, 4, 5 are different enough for reads to get assigned independently.

probability_red = configuration_red* red_count[i] /(red_count[i]+green_count[i])
To determine the probability of each configuration for the colour exon assignments, a probability for the red exons (**probability_red**) is calculated using the formula. Below formula accounts for each exon by adjusting for the relative frequency of reads mapped to red and green counts, allowing the assignment of probabilities to configurations in a manner that distinguishes between red and green exon reads. **probability_green** is just complementary product of the two terms.

Functions elaboration

Data Loading and Preparation

- Loads the BWT last column data and index mapping data from specified file paths.
- Reads the last column of the BWT as chunks, removes newline characters, and stores it in a NumPy array as single-byte elements.
- Reads the mapping file, selecting rows at intervals defined by delta, creating a downsampled index mapping. This mapping allows retrieval of chromosome positions based on BWT ranks.

Boundary Adjustment and Milestone Matrix Creation

- Adjusts the upper and lower bounds for a character in the BWT column based on given milestones and delta, allowing a compressed search within the BWT.
- Constructs a matrix to store cumulative counts of nucleotide occurrences (A, C, G, T) up to each milestone based on delta intervals.

- Generates masks for each nucleotide and updates cumulative counts across the milestone matrix, aiding in efficient rank queries.

Rank and Select Queries for Character Mapping

- Uses milestone data to count occurrences of a character up to a given index, allowing faster lookups within the BWT column by leveraging milestone intervals.
- Returns the exact position of a specified character at a particular rank in the BWT by using cumulative counts of each nucleotide.

Locating Positions for Reads

- Locates sequences in the BWT column using partial scans and adjusted boundaries for compressed searching.
- Returns a list of positions where the read aligns in the genome by iterating over BWT positions.

Processing Reads and Counting Exon Occurrences

- Reads sequences from the data file and performs BWT-based location searches.
- For each read, checks if it contains a target sequence pattern, maps positions using locate_reads, and stores matched indices for specified genomic ranges.
- Counts the occurrences of read positions that fall within specified exon ranges (e.g., red_exons and green_exons).

Computing probability

- This calculates the probability of occurrence of the row of configuration.

Output

The test read aligns at position 149249814.
Gene start positions: [149249757, 149288166, 149325284]
configuration_red = [[0.5, 0.5, 0.5, 0.5], [1.0, 1.0, 0.0, 0.0], [0.33, 0.33, 1.0, 1.0], [0.33, 0.33, 0.33, 1.0]]
probability_red = [0.0005594912220112581, 0.0, 0.0009748575052324162, 0.00032170297672669734]
As configuration 3 has the highest probability of occurrence, it is the most likely configuration for colour blindness.
red_counts = [256, 57, 42.5, 100, 203.5, 285]
green_counts = [256, 193.5, 132.5, 133.5, 334.5, 285]